Kimberly Wang

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Education

Rensselaer Polytechnic Institute Troy, NY
B.S. Electrical Engineering, GPA: 3.71 *August 2024 – Present*
Relevant Coursework: Electric Circuits, Computer Components and Operations, Embedded Control, Intro to Logic Based AI, Intro to Electronics (Spring 2026), Signals & Systems (Spring 2026), Engineering Probability (Spring 2026)

Technical Skills

Hardwares: Arduino, MSP432, Basys 3, Soldering, Oscilloscope, Waveform Generator, Multimeter, Breadboard
Softwares: LTspice, Cadence, Altium Designer, Vivado, Siemens NX, LabVIEW, MATLAB, Simulink, Office Suite
Languages: Python, C, VHDL

Experience

Undergraduate Researcher @ RPI Mercer XLab Jan 2025 – Present
Rensselaer Polytechnic Institute *Troy, NY*
Software Configured Breadboard (SCB) Supervised by ADI Fellow Douglas Mercer

- Use python and ALICE software to computerize the wiring of breadboards to reduce debugging and wiring issues
- Verify the functionality of the SCB using scripts and diagnose errors with a multimeter
- Write user guides to using and setting up the SCB and help students with troubleshooting problems
- Program in Python to make improvements to firmware and features
- Collaborating with RPai (RPI's AI club) to integrate an AI agent to automate circuit design and testing

Undergraduate Assistant for "Intro to ECSE" Aug. 2025 – Present
Rensselaer Polytechnic Institute *Troy, NY*

- Help students in debugging and implementing C programming on Arduino and MATLAB
- Advise on analog and digital circuit building, testing, and measurement
- Clarify questions about theoretical concepts within contexts of labs

Projects

CO Sensing and Ventilating System | *LTspice, Breadboard* Sep 2025 – Present

- Created a multi-stage system including CO sensing, signal filtering, and ventilation controls
- Designed and built the initial indication stage using an RC filter and comparator with an LED output
- Configure an integrator to ramp from 0 - 5V in 10s when triggered on and when off, reset to 0V in 2s
- Use a 555 Timer to do PWM on the ramping voltage signal with an adjustable duty cycle
- Currently working on implementation of final stage

Compass Controlled TI RSLK Car | *MSP432, C, I2C* Apr 2025

- Use a compass to instruct a to RSLK Car Movements
- C programming GPIO of the MSP432 microcontroller to interact with a magnetic sensor using I2C protocols to get data for directions
- Implement PID control for the motor movements based on instruction

Vending Machine Moore FSM | *Logisim, Breadboard* Apr 2025

- Designed a multi-state Moore FSM taking inputs of "coins" to reach a specific total to "vend" an item
- Simulated on Logisim to validate state transitions based on inputs and output
- Construct and troubleshoot a circuit built with 74x series chips to implement the design

Implementation of ALU to Demonstrate Boolean Functions | *Vivado, VHDL, Basys 3* Feb 2025

- Authored VHDL code to write logic for an ALU with 4 bits for an adder, subtractor, bit parity, and comparator
- Used Vivado to define constraints and generate bitstream to be sent to Basys 3 FPGA board
- Tested functions with on-board switches as inputs and LEDs for outputs

PI Controller Design | *LTspice, Breadboard* Nov 2024

- Design and simulate an analog PI controller in LTspice using two inverting op amps and a parallel RC circuit to achieve a steady state voltage equal to input by 3s
- Verified the design by building the circuit to get results with $< 5\%$ error